

3.2 Jobs Performed a Few Times per Shift

3.2.1 Loading Punch Press Stock, Example 1

3.2.1.1 Job Description

Figure 5 illustrates a common oversight in physically stressful jobs. A punch press operator routinely handles small parts, feeding them into a press and removing them. A cursory view of this task may overlook the fact that once per shift the operator must load a heavy reel of supply stock (illustrated at floor height) from the floor onto the machine. The diameter of the reel is 30 inches, the width of the reel between the worker's hands is 12 inches, and the reel weighs 44 lbs. significant control of the load is required at the destination of the lift due to the design of the machine. Also, the worker cannot get closer to the roll (i.e., between the legs) because the roll is too awkward.

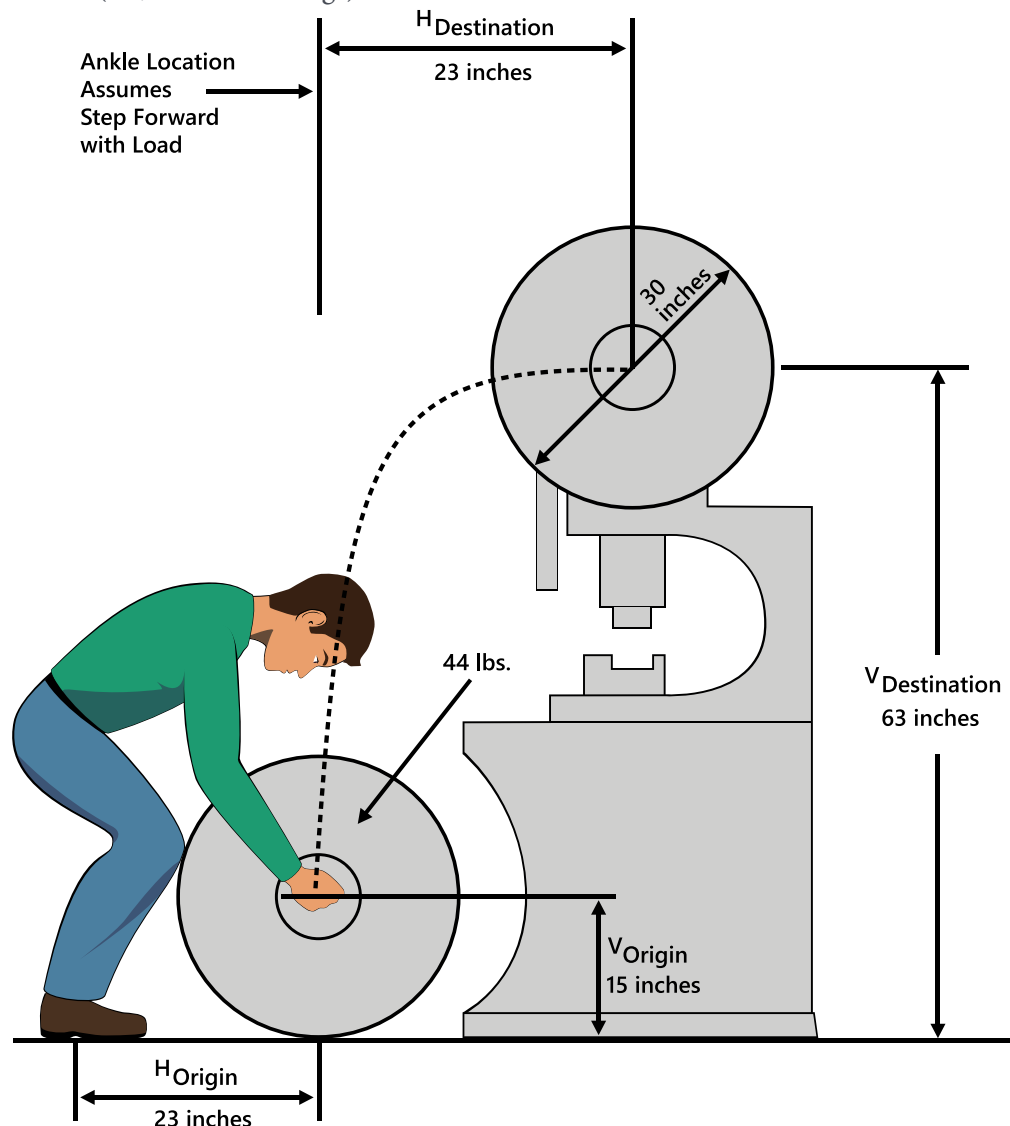


Figure 5: Loading Punch Press Stock, Example 1

3.2.1.2 Job Analysis

The task variable data are measured and recorded on the job analysis worksheet (Figure 6). Assuming the operator lifts the reel in the plane shown, rather than on the side of the machine, the vertical height (V) at the origin is 15 inches, the vertical height (V) at the destination is 63 inches, and the horizontal distance (H) is 23 inches at both the origin and the destination of the lift. The activity occurs only once per shift, so F is assumed to be <0.2 (see Table 5), and duration is assumed to be less than 1 hour.

No asymmetric lifting is involved (i.e., $A=0$), and according to Table 6, the couplings are classified as fair because the object is irregular and the fingers can be flexed about 90 degrees. Since significant control is required at the destination, the RWL must be computed at both the origin and the destination of the lift.

The multipliers are determined from the lifting equation or from tables (Tables 1 to 5, and Table 7). The CM is .95 at the origin and 1.0 at the destination, due to the difference in the vertical height at the origin and destination. As shown in Figure 6, the RWL for this activity is 16.3 lbs at the origin and 14.5 lbs at the destination.

3.2.1.3 Hazard Assessment

The weight to be lifted (44 lbs) is greater than the RWL at both the origin and the destination of the lifts (16.3 lbs and 14.5 lbs, respectively). The LI at the origin is $44/16.3$ or 2.7, and the LI at the destination is $44/14.5$ or 3.0. These values indicate that this lift would be hazardous for a majority of healthy industrial workers.

3.2.1.4 Redesign Suggestions

The worksheet shown in Figure 6 indicates that the smallest multipliers (i.e., the greatest penalties) are .44 for the HM, .75 for the VM at the destination, and .86 for the DM. Using Table 8, the following job modifications are suggested:

1. Bring the object closer to the worker at the destination to increase the HM value.
2. Lower the destination of the lift to increase the VM value.
3. Reduce the vertical travel distance between the origin and the destination of the lift to increase the DM value.
4. Modify the job so that significant control of the object at the destination is not required. This will eliminate the need to use the lower RW value at the destination.

If the operator could load the machine from the side, rather than from the front, the reel could be turned 90° which would reduce the horizontal location of the hands at the origin (i.e., $H = 10$ inches) and destination of the lift (i.e., $H = 12$ inches).

The grip, however, would be poor because the object is bulky and hard to handle and the fingers could not be flexed near 90° when picking up the reel (see Table 6, Note 4).

The RWL and corresponding LI values for this preferred combination of task variables (i.e., loading the machine from the side) are shown on the modified job analysis sheet (Figure 7). At the origin, the RWL is 35.1 lbs and the LI is 1.3. At the destination, the RWL is 24.6 lbs and the LI is 1.8. Since the LI is still greater than 1.0, however, a more comprehensive

solution may be needed. This could include: (1) lowering the vertical height of the destination, which would increase the VM and the DM at both the origin and the destination of the lift; (2) reducing the size and/or weight of the supply reel; or, (3) transferring the supply reel from the storage area on a mobile, mechanical lifting device or jack that could be moved near the machine to eliminate the need for manual lifting. If it is not feasible to eliminate or redesign the job, then other measures, such as assigning two or more workers, could be considered as an interim control procedure.

3.2.1.5 Comments

Although ergonomic redesign is preferred, this example demonstrates how a change in work practices (i.e., insuring that the operator can load the reel from the side) can reduce the magnitude of physical stress associated with a manual lifting task. This approach, however, relies more on worker compliance than on physical job modifications.

3.2.2 Loading Supply Rolls, Example 2

3.2.2.1 Job Description

With both hands directly in front of the body, a worker lifts the core of a 35-lb roll of paper from a cart, and then shifts the roll in the hands and holds it by the sides to position it on a machine, as shown in Figure 8. Significant control of the roll is required at the destination of the lift. Also, the worker must crouch at the destination of the lift to support the roll in front of the body, but does not have to twist.

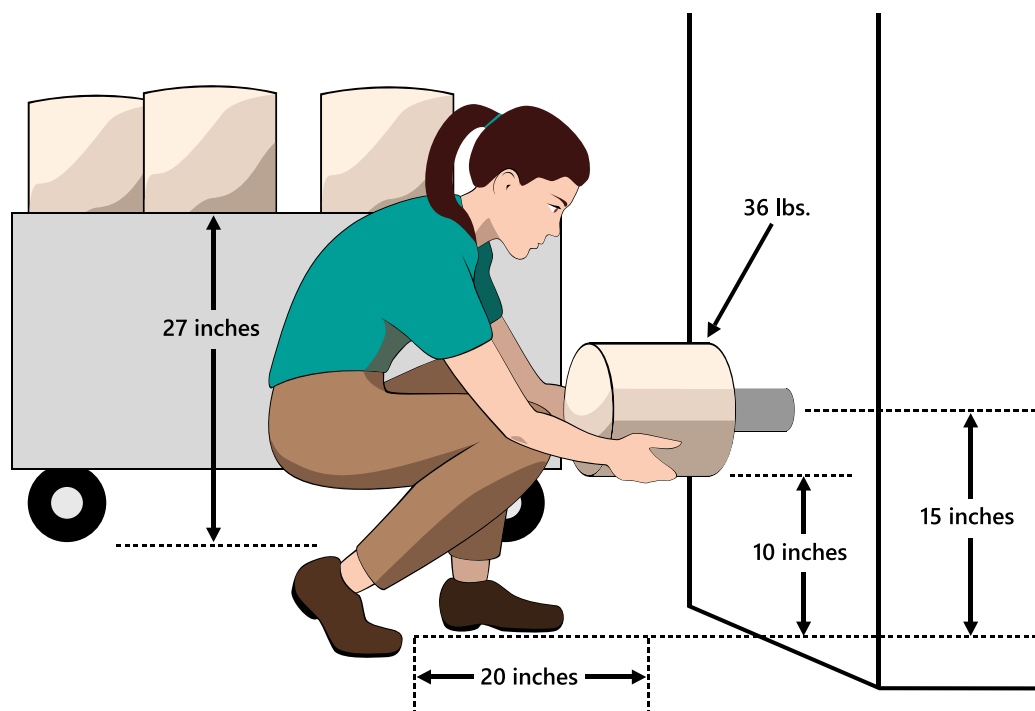


Figure 8: Loading Supply Rolls, Example 2

3.2.2.2 Job Analysis

The task variable data are measured and recorded on the job analysis worksheet (Figure 9). The vertical location of the hands is 27 inches at the origin and 10 inches at the destination. The horizontal location of the hands is 15 inches at the origin and 20 inches at the destination. The asymmetry angle is 0 degrees at both the origin and the destination and the frequency is 4 lifts/shift (i.e., less than .2 lifts/min for less than 1 hour – see Table 5).

Using Table 6, the coupling is classified as poor because the worker must reposition the hands at the destination of the lift and they cannot flex the fingers to the desired 90° angle (e.g., hook grip). No asymmetric lifting is involved (i.e., A=0), and significant control of the object is required at the destination of the lift. Thus, the RWL should be computed at both the origin and the destination of the lift. The multipliers are computed from the lifting equation or determined from the multiplier tables (Tables 1 to 5, and Table 7). As shown in Figure 9, the RWL for this activity is 28.0 lbs at the origin and 18.1 lbs at the destination.

3.2.2.3 Hazard Assessment

The weight to be lifted (35 lbs) is greater than the RWL at both the origin and destination of the lift (28.0 lbs and 18.1 lbs, respectively). The LI at the origin is 35 lbs/28.0 lbs or 1.3, and the LI at the destination is 35 lbs/18.1 lbs or 1.9. These values indicate that this job is only slightly stressful at the origin, but moderately stressful at the destination of the lift.

3.2.2.4 Redesign Suggestions

The first choice for reducing the risk of injury for workers performing this task would be to adapt the cart so that the paper rolls could be easily pushed into position on the machine, without manually lifting them.

If the cart cannot be modified, then the results of the equation may be used to suggest task modifications. The worksheet displayed in Figure 9 indicates that the multipliers with the smallest magnitude (i.e., those providing the greatest penalties) are .50 for the HM at the destination, .67 for the HM at the origin, .85 for the VM at the destination, and .90 for the CM value. Using Table 8, the following job modifications are suggested:

1. Bring the load closer to the worker by making the roll smaller so that the roll can be lifted from between the worker's legs. This will decrease the H value, which in turn will increase the HM value.
2. Raise the height of the destination to increase the VM.
3. Improve the coupling to increase the CM.

If the size of the roll cannot be reduced then the vertical height (V) of the destination should be increased. Figure 10 shows that if V was increased to about 30 inches, then VM would be increased from .85 to 1.0; the H value would be decreased from 20 inches to 15 inches, which would increase HM from .50 to .67; the DM would be increased from .93 to 1.0. Thus, the final RWL would be increased from 18.1 lbs to 30.8 lbs, and the LI at the destination would decrease from 1.9 to 1.1.

In some cases, redesign may not be feasible. In these cases, use of a mechanical lift may be more suitable. As an interim control strategy, two or more workers may be assigned to lift the supply roll.

3.2.2.5 Comments

The horizontal distance (H) is a significant factor that may be difficult to reduce because the size of the paper rolls may be fixed. Moreover, redesign of the machine may not be practical. Therefore, elimination of the manual lifting component of the job may be more appropriate than job redesign.

3.2.3 Loading Bags into a Hopper, Example 3

3.2.3.1 Job Description

The worker positions himself midway between the hand truck and the mixing hopper, as illustrated in Figure 11. Without moving his feet he twists to the right and picks up a bag off the hand truck. In one continuous motion he then twists to his left to place the bag on the rim of the hopper. A sharp edged blade within the hopper cuts open the bag to allow the contents to fall into the hopper. This task is done infrequently (i.e., 1–12 times per shift) with large recovery periods between lifts (i.e., >1.0 Recovery Time/Work Time ratio). In observing the worker perform the job, it was determined that the non-lifting activities could be disregarded because they require minimal force and energy expenditure.

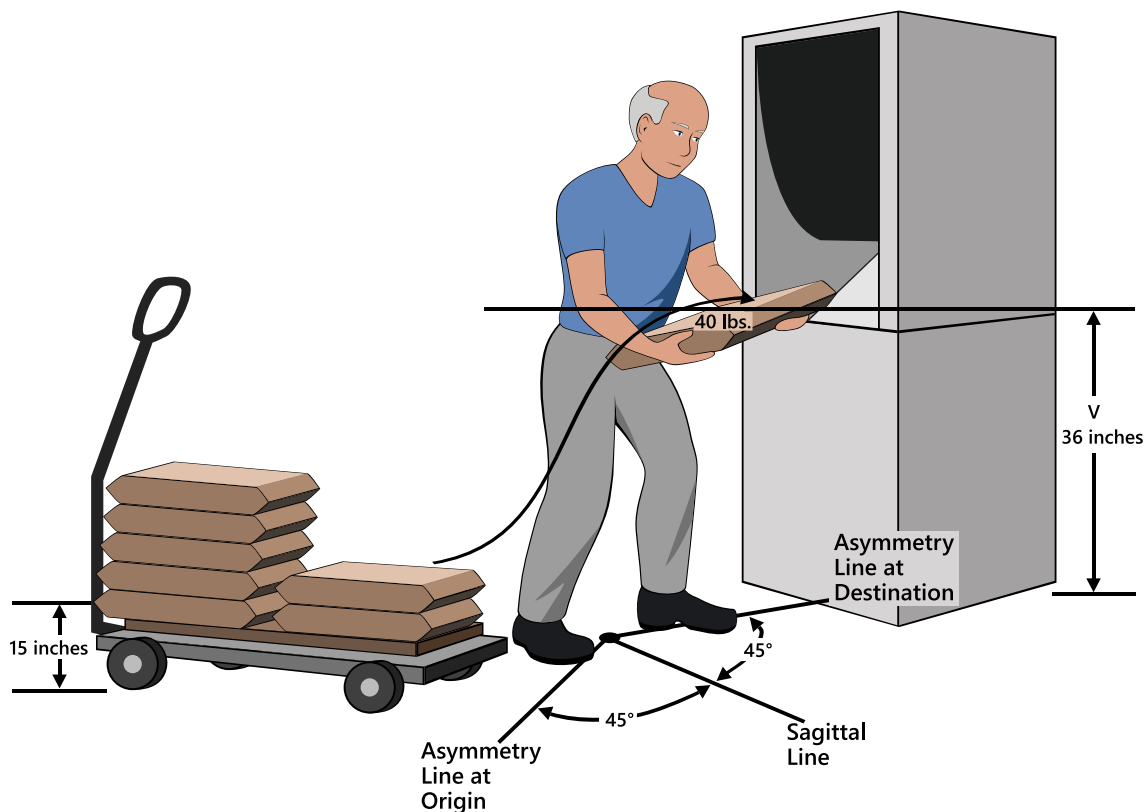


Figure 11: Loading Bags Into Hopper, Example 3

Significant control is not required at the destination but the worker twists at the origin and destination of the lift. Although several bags are stacked on the hand truck, the highest risk of overexertion injury is associated with the bag on the bottom of the stack; therefore only the lifting of the bottom bag will be examined. Note, however, that the frequency multiplier is based on the overall frequency of lifting for all of the bags.

3.2.3.2 Job Analysis

The task variable data are measured and recorded on the job analysis worksheet (Figure 12). The vertical location of the hands is 15 inches at the origin and 36 inches at the destination. The horizontal location of the hands is 18 inches at the origin and 10 inches at the destination. The asymmetry angle is 45° at the origin and 45° at the destination of the lift and the frequency is less than .2 lifts/min for less than 1 hour (see Table 5).

Using Table 6, the coupling is classified as fair because the worker can flex the fingers about 90° and the bags are semi-rigid (i.e., they do not sag in the middle). Significant control of the object is not required at the destination of the lift so the RWL is computed only at the origin. The multipliers are computed from the lifting equation or determined from the multiplier tables (Tables 1 to 5, and Table 7). As shown in Figure 12, the RWL for this activity is 18.9 lbs.

3.2.3.3 Hazard Assessment

The weight to be lifted (40 lbs) is greater than the RWL (18.9 lbs). Therefore, the LI is 40/18.9 or 2.1. This job would be physically stressful for many industrial workers.

3.2.3.4 Redesign Suggestions

The worksheet shows that the smallest multipliers (i.e., the greatest penalties) are .56 for the HM, .86 for the AM and .89 for the VM. Using Table 8, the following job modifications are suggested:

1. Bringing the load closer to the worker to increase the HM.
2. Reducing the asymmetry angle to increase AM. This could be accomplished either by moving the origin and destination points closer together or further apart.
3. Raising the height at the origin to increase the VM.

If the worker could get closer to the bag before lifting the H value could be decreased to 10 inches, which would increase the HM to 1.0, the RWL would be increased to 33.7 lbs, and the LI would be decreased to 1.2 (i.e., 40/33.7).

3.2.3.5 Comments

This example demonstrates that certain lifting jobs may be evaluated as a single-task or multi-task job. In this case, only the most stressful component of the job was evaluated. For repetitive lifting jobs, the multi-task approach may be more appropriate. (See Examples 7–10).

3.3 Single Task, Performed Repetitively

3.3.1 Package Inspection, Example 4

3.3.1.1 Job Description

The job illustrated in Figure 13 consists of a worker inspecting compact containers for damage on a low shelf, and then lifting them with both hands directly in front of the body from shelf 1 to shelf 2 at a rate of 3/min for a duration of 45 minutes. For this analysis, assume that (1) the worker cannot take a step forward when placing the object at the destination, due to the bottom shelf, and (2) significant control of the object is required at the destination. The containers are of optimal design, but without handles. (For classification, refer to Table 6).

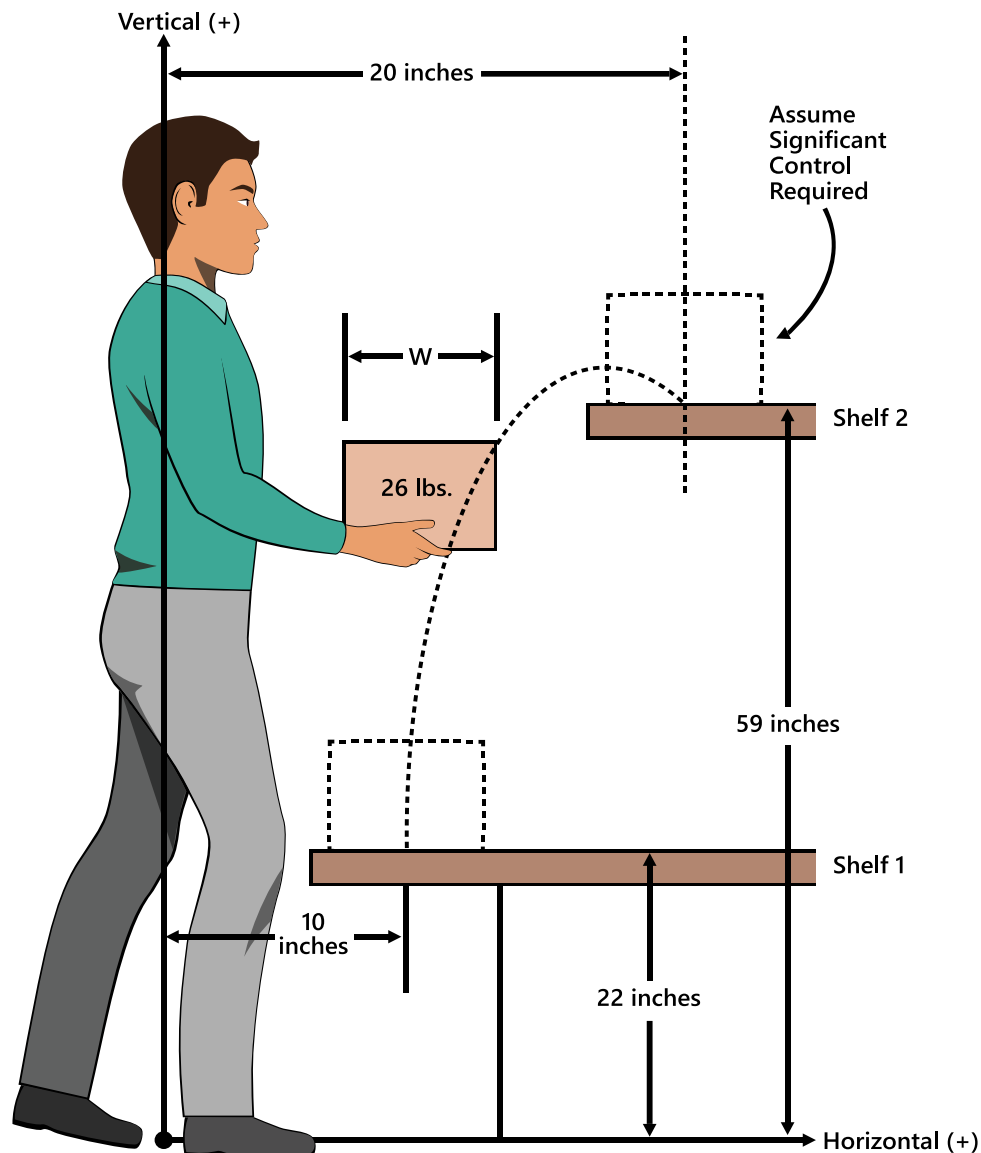


Figure 13: Package Inspection, Example 4

3.3.1.2 Job Analysis

The task variable data are measured and recorded on the task analysis worksheet (Figure 14). The horizontal distance at the origin of the lift is 10 inches and the horizontal distance at the destination of the lift is 20 inches. The height of shelf one is 22 inches and the height of shelf two is 59 inches. Since the container is of optimal design but does not have handles or handhold cutouts, the coupling is defined as “fair” (see Table 6). No asymmetric lifting is involved (i.e., $A=0$). Significant control of the load is required at the destination of the lifts. Therefore the RWL is computed at both the origin and the destination of the lift.

The multipliers are computed from the lifting equation or determined from the multiplier tables (Tables 1 to 5, and Table 7). As shown in Figure 14, the RWL for this activity is 34.9 lbs at the origin and 15.2 lbs at the destination.

3.3.1.3 Hazard Assessment

The weight to be lifted (26 lbs) is less than the RWL at the origin (34.9 lbs) but greater than the RWL at the destination (15.2 lbs). The LI is $26/34.9$ or .76 (rounded to .8) at the origin, and the LI is $26/15.2$ or 1.7 at the destination. These values indicate that the destination of the lift is more stressful than the origin, and that some healthy workers would find this task physically stressful.

3.3.1.4 Redesign Suggestions

The worksheet illustrated in Figure 14 shows that the multipliers with the smallest magnitude (i.e., those that provide the greatest penalties) are .50 for the HM at the destination, .78 for the VM, .87 for the DM and .88 for the FM at the destination of the lift. Using Table 8, the following job modifications are suggested:

1. Bring the destination point closer to the worker to increase the HM value.
2. Lower the height of shelf 2 to increase the VM value.
3. Decrease the vertical distance between origin and destination of lift to increase the DM value.
4. Reduce the lifting frequency rate to increase the FM value.
5. Modify the task so that there is no need for significant control of the object at the destination to eliminate the lower RWL value.

Practical job modifications could include bringing shelf 2 closer to the worker to reduce H, raising the height of shelf 1 to increase the CM value, lowering the height of shelf 2 to reduce D, or reducing the need for significant control at the end of the lift by providing a receiving chute.

3.3.1.5 Comments

Since the lifting pattern is continuous over the 45 minute work session, the lifting frequency is not adjusted using the special procedure.

3.3.2 Dish-Washing Machine Unloading, Example 5

3.3.2.1 Job Description

A worker manually lifts trays of clean dishes from a conveyor at the end of a dishwashing machine and loads them on a cart as shown in Figure 15. The trays are filled with assorted dishes (e.g., glasses, plates, bowls) and silverware. The job takes between 45 minutes and 1 hour to complete and the lifting frequency rate averages 5 lifts/min. Workers usually twist to one side of their body to lift the trays (i.e., asymmetric lift) and then rotate to the other side of their body to lower the trays to the cart in one smooth continuous motion. The maximum amount of asymmetric twist varies between workers and within workers, however, there is usually equal twist to either side. During the lift the worker may take a step toward the cart. The trays have well-designed handheld cutouts and are made of lightweight materials.

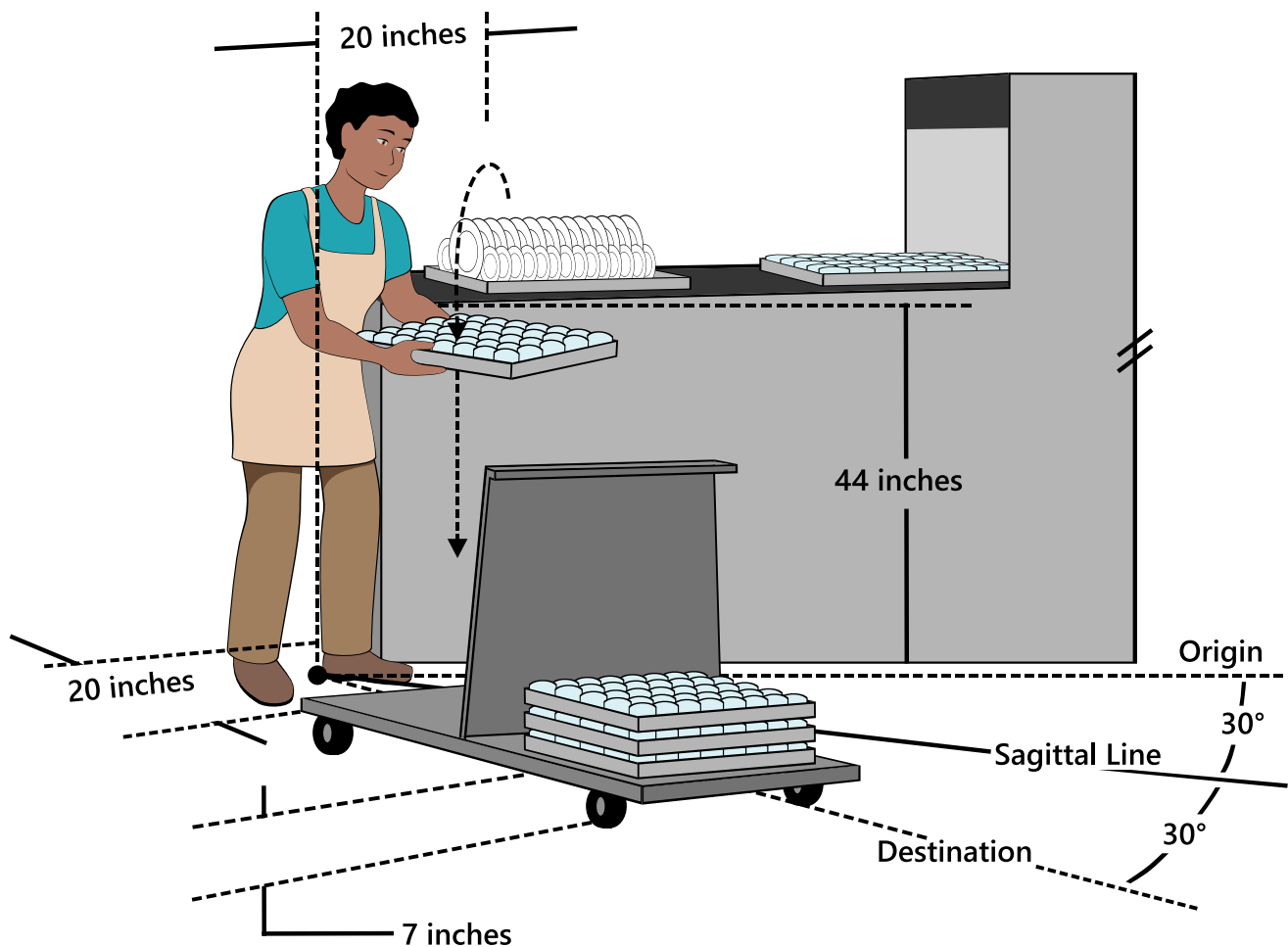


Figure 15: Dishwashing Machine Unloading, Example 5

3.3.2.2 Job Analysis

The task variable data are measured and recorded on the job analysis worksheet (Figure 16). At the origin of the lift the horizontal distance (H) is 20 inches, the vertical distance (V) is 44 inches, and the asymmetry angle (A) is 30°. At the destination of the lift, H is 20 inches, V is 7 inches, and A is 30°. The trays normally weigh from 5 lbs to 20 lbs, but for this example, assume that all of the trays weigh 20 lbs.

Using Table 6, the coupling is classified as **Good**. Significant control is required at the destination of the lift. Using Table 5, the FM is determined to be .80. As shown in Figure 16, the RWL is 14.4 lbs at the origin and 13.3 lbs at the destination.

3.3.2.3 Hazard Assessment

The weight to be lifted (20 lbs) is greater than the RWL at both the origin and destination of the lift (14.4 lbs and 13.3 lbs, respectively). The LI at the origin is 20/14.4 or 1.4 and the LI at the destination is 1.5. These results indicate that this lifting task would be stressful for some workers.

3.3.2.4 Redesign Suggestions

The worksheet shows that the smallest multipliers (i.e., the greatest penalties are .50 for the HM, .80 for the FM, .83 for the VM, and .90 for the AM. Using Table 8, the following job modifications are suggested:

1. Bring the load closer to the worker to increase HM.
2. Reduce the lifting frequency rate to increase FM.
3. Raise the destination of the lift to increase VM.
4. Reduce the angle of twist to increase AM by either moving the origin and destination closer together or moving them further apart. Since the horizontal distance (H) is dependent on the width of the tray in the sagittal plane, this variable can only be reduced by using smaller trays. Both the DM and VM, however, can be increased by lowering the height of the origin and increasing the height of the destination. For example if the height at both the origin and destination is 30 inches, then VM and DM are 1.0, as shown in the modified worksheet (Figure 17). Moreover, if the cart is moved so that the twist is eliminated, the AM can be increased from .90 to 1.00. As shown in Figure 17, with these redesign suggestions, the RWL can be increased from 13.3 lbs to 20.4 lbs, and the LI values are reduced to 1.0.

3.3.2.5 Comments

This analysis was based on a one-hour work session. If a subsequent work session begins before the appropriate recovery period has elapsed, then the two-hour category would be used to compute the FM value.

As in the previous example, since the lifting pattern is continuous over the full duration of the work sample (i.e., more than 15 minutes), the lifting frequency is not adjusted using the special procedure.

3.3.3 Product Packaging 1, Example 6

3.3.3.1 Job Description

In the job illustrated in Figure 18, products weighing 25 lbs arrive via a conveyor at a rate of 1-per minute where a worker packages the product in a cardboard box and then slides the packaged box to a conveyor behind table B. Assume that significant control of the object is not required at the destination, but that the worker twists to pick up the product; also assume that the worker can flex the fingers to the desired 90° angle to grasp the container. The job is performed for a normal 8-hour shift, including regular rest allowance breaks.

3.3.3.2 Job Analysis

The task variable data are measured and recorded on the job analysis worksheet (Figure 19). At the origin, the vertical location (V) is 24 inches and the horizontal location is 14 inches. At the destination the vertical location is 40 inches, which represents the height of table B plus the height of the box, and the horizontal location is 16 inches.

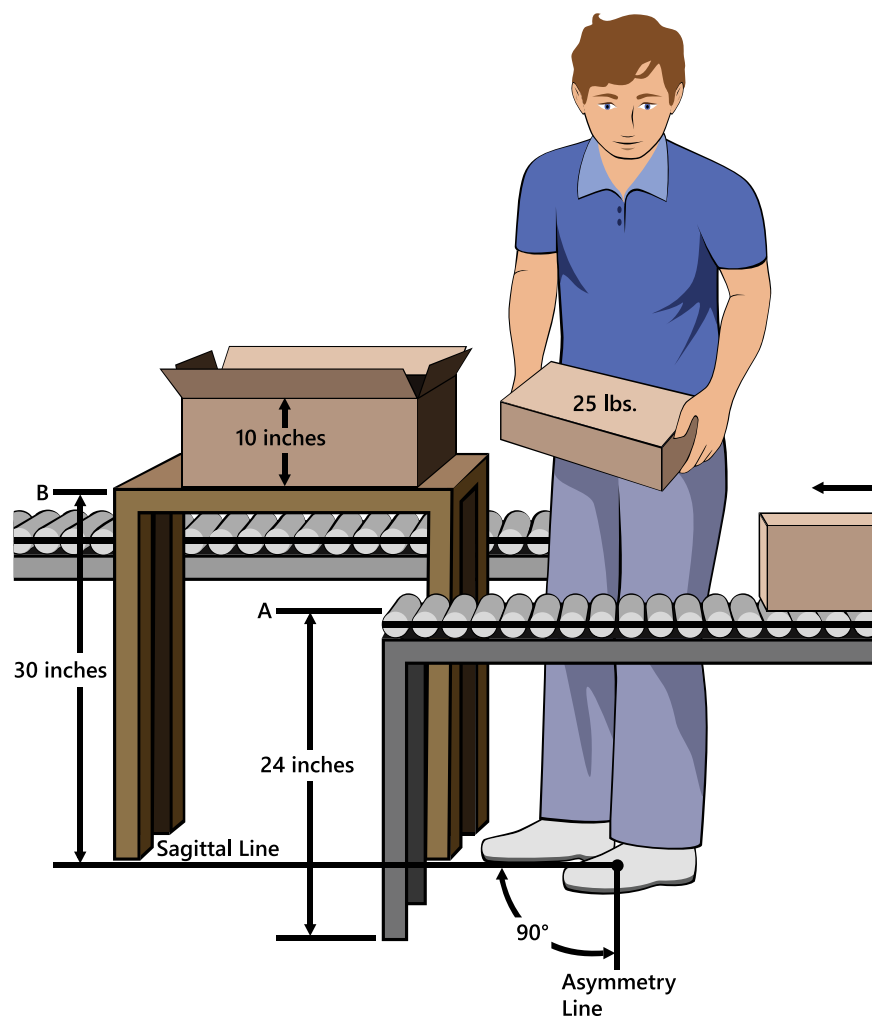


Figure 18: Product Packaging I, Example 6

Using Table 6, the coupling is classified as **fair**. The worker twists 90° to pick up the product. The job is performed for an 8-hour shift with a frequency rate of 1-lift per minute. Using Table 5, the FM is determined to be .75. Since significant control is not required at the destination then the RWL is only computed at the origin of the lift. The multipliers are computed from the lifting equation or determined from the multiplier tables (Tables 1 to 5, and Table 7). As shown in Figure 19, the RWL for this lifting task is 16.4 lbs.

3.3.3.3 Hazard Assessment

The weight to be lifted (25 lbs) is greater than the RWL (16.4 lbs). Therefore, the LI is 25/16.4 or 1.5. This task would be stressful for some healthy workers.

3.3.3.4 Redesign Suggestions

The worksheet shows that the multipliers with the smallest magnitude (i.e., those providing the greatest penalties) are .71 for the HM, .71 for the AM, .75 for the FM. Using Table 8, the following job modifications are suggested:

1. Bring the load closer to the worker to increase HM.
2. Move the lift's origin and destination closer together to reduce the angle of twist and increase the AM.
3. Reduce the lifting frequency rate and/or provide longer recovery periods to increase FM.

Assuming that the large horizontal distance is due to the size of the object lifted rather than the existence of a barrier, then the horizontal distance could only be reduced by making the object smaller or re-orienting the object. An alternate approach would be to eliminate body twist by providing a curved chute to bring the object in front of the worker. For this modified job (worksheet shown in Figure 20), the AM is increased from 0.71 to 1.0, the HM is increased from 0.71 to 0.77, the RWL is increased from 16.4 lbs to 25 lbs, and the LI is decreased from 1.5 to 1.00. Eliminating body twist reduces the physical stress to an acceptable level for most workers. Alternate redesign recommendations could include: (1) raising the height of conveyor A and/or reducing the height of work bench B; or, (2) providing good couplings on the containers. For example, the curved chute could also be designed to bring the load to a height of 30 inches. This would increase the VM, DM, and CM values to 1.0, which would reduce the lifting index even further.

3.3.3.5 Comments

Although several alternate redesign suggestions are provided, reducing the asymmetry angle should be given a high priority because a significant number of overexertion lifting injuries are associated with excessive lumbar rotation and flexion.

As in the earlier examples, the lifting pattern is continuous over the full duration of the work sessions. Thus, the lifting frequency is not adjusted using the special procedure described in the Frequency Component section.

3.4 Repetitive Multi-Task, Short Duration

3.4.1 Depalletizing Operation, Example 7

3.4.1.1 Job Description

A worker unloads 12lb cartons from a pallet onto a conveyor as illustrated in Figure 21. The cartons are vertically stacked from the floor in five tiers. No twisting is required when picking up and putting down the cartons, and the worker is free to step on each pallet to get close to the carton (i.e., only one layer in depth from the front of the pallet must be analyzed). Walking and carrying are minimized by keeping the pallet close to the conveyor and significant control of the object is not required at the destination of the lift. The vertical location (V) at the origin, horizontal location (H), and vertical travel distance (D), vary from one lift to the next.



Figure 21: Depalletizing Operation, Example 7

3.4.1.2 Job Analysis

Since the job consists of more than one distinct task and the task variables often change, the multi-task lifting analysis procedure should be used.

This job is divided into five tasks representing the five tiers of loaded pallets. Task numbering is arbitrary and the sequencing does not reflect the order in which the tasks are performed. It is important, however, to identify each distinct type of lifting task. **Note, it may not be appropriate to use the lifting equation for mixed-task jobs that require significant amounts of pushing, pulling or carrying.**

The following measurements/observations were made and recorded on the job analysis worksheet (Figure 22):

1. Carton dimensions are 16 inches × 16 inches × 16 inches.
2. The vertical locations at the origin represent the position of the hands under the cartons. The top of the conveyor is 20 inches from the floor.
3. For this example, assume that the horizontal locations were not measured, but estimated using the formulas provided in the Horizontal Multiplier section. From these formulas, $H = (8 + 16/2)$ or 16 inches for the top four tiers and $H = (10 + 16/2)$ or 18 inches for the bottom tier.
4. The pallet is 4 inches in height.
5. No asymmetric lifting is involved (i.e., $A=0$).
6. Cartons are continuously unloaded at the rate of 12-per minute (i.e., 2.4 lifts/min per tier) for 1 hour.
7. The job consists of continuous 1-hour work sessions separated by 90-minute recovery periods.
8. Using Table 6, the coupling is classified as fair.

The multi-task lifting analysis consists of the following three steps:

1. Compute the frequency-independent- RWL (FIRWL) and frequency-independent-lifting index (FILI) values for each task using a default FM of 1.0.
2. Compute the single-task- RWL (STRWL) and single-task-lifting index (STLI) for each task. **Note, in this example, interpolation was used to compute the FM value for each task because the lifting frequency rate was not a whole number (i.e., 2.4).**
3. Renumber the tasks in order of decreasing physical stress as determined from the STLI value, starting with the task with the largest STLI.

Compute the FIRWL and the FILI values for each task using a default FM of 1.0. The multi-task lifting analysis consists of the following three steps:

Step 1

Compute the frequency-independent-RWL (FIRWL) and frequency-independent-lifting index FILI values for each task using a default FM of 1.0.

	FIRWL	FILI
Tier 1	20.4 lbs	.6
Tier 2	28.4 lbs	.4
Tier 3	28.7 lbs	.4
Tier 4	23.8 lbs	.5
Tier 5	19.9 lbs	.6

These results indicate that none of the tasks are particularly stressful, from a strength point of view, but that tiers 1 and 5 do require the most strength. Remember, however, that these results do not take the frequency of lifting into consideration.

Step 2

Compute the STRWL and STLI values for each task, where $STRWL = FIRWL \times FM$. The FM for each task is determined by interpolating between the FM values for 2 and 3 lifts/minute from Column 2 of Table 5. The results are displayed in Figure 22.

	STRWL	STLI
Tier 1	18.4 lbs	.7
Tier 2	25.6 lbs	.5
Tier 3	25.8 lbs	.5
Tier 4	21.4 lbs	.6
Tier 5	17.9 lbs	.7

These results suggest that none of the tasks are stressful, if performed individually. Note, however, that these values do not consider the combined effects of all of the tasks.

Step 3

Renumber the tasks starting with the task with the largest STLI value, and ending with the task with the smallest STLI value. If more than one task has the same STLI value, assign the lower task number to the task with the highest frequency.

3.4.1.3 Hazard Assessment

Compute the composite-lifting index (CLI) for the job, using the numbered tasks as described in the Multi-Task procedure.

As shown on Figure 22, the CLI value for this job is 1.4. This means that some healthy workers would find this job physically stressful. Therefore, some redesign may be needed. Analysis of the results suggest that any three of these tasks would probably result in a CLI below 1.0, which would be acceptable for nearly all healthy workers. However, when the other two tasks are added, the overall frequency increases the lifting index above 1.0. This suggests that the overall frequency should be reduced to limit the physical stress associated with this job.

3.4.1.4. Redesign Suggestion

The worksheet illustrated in Figure 22 indicates that the multipliers with the smallest magnitude (i.e., those providing the greatest penalties) are .56 for the HM at tier 1; .63 for the HM at tiers 2 through 5; .72 for the VM at tier 5; and .81 for the VM at tier 1. Using Table 8, the following job modifications are suggested:

1. Bring the cartons closer to the worker to increase the HM value.
2. Lower the height for tier five to increase the VM value.
3. Raise the height of tier one to increase the VM value.

The FILI values are all less than 1.0, indicating that strength should not be a problem for any of these tasks. Moreover, the STLI were all less than 1.0 indicating the one of the tasks would be physically stressful, if performed individually. When the combined physical demands of the tasks are considered, however, the resulting CLI exceeds 1.0. This is likely due to the high frequency rate for the combined job. Since a number of simplifying assumptions were made in this example, however, a more detailed metabolic analysis of such a job may be needed before implementing ergonomic redesign. Such an analysis is described in detail by Garg *et al.* (1978).

An engineering approach should be the first choice for job redesign (i.e., physical changes in layout; such as raising or lowering shelves, tables, or pallets) rather than worker compliance. In this case the high frequency rate is a significant problem and should be reduced. A reduction in frequency could decrease the CLI to about 1.0.

3.4.1.5 Comments

With more complicated tasks, such a simple solution will not necessarily be possible, and more detailed analyses may be required to determine compressive forces, strength requirements, and energy expenditures.

3.4.2 Handling Cans of Liquid, Example 8

3.4.2.1 Job Description

A worker unloads cans of liquids from a cart to three storage shelves as shown in Figure 23. Although the cans are lifted in the sagittal plane when moved between shelves, they are usually lifted asymmetrically from one side of the body to the other when lifted from the cart to the shelves. The worker may take a step when placing the cans onto the shelf. The cans do not have molded handholds, so the worker hooks his fingers or slides his hand under the turned edge of the can to lift. When lifting to the top shelf, workers usually reposition their grip near the end of the lift. The work pattern consists of intermittent, six-minute work sessions separated by three-minute recovery periods. The actual lifting frequency during the six-minute work sessions was 9 lifts/minute. There is a 90-minute break after each hour of work.

3.4.2.2 Job Analysis

Since the job consists of more than one distinct task and the task variables change often, the multi-task lifting analysis procedure should be used.

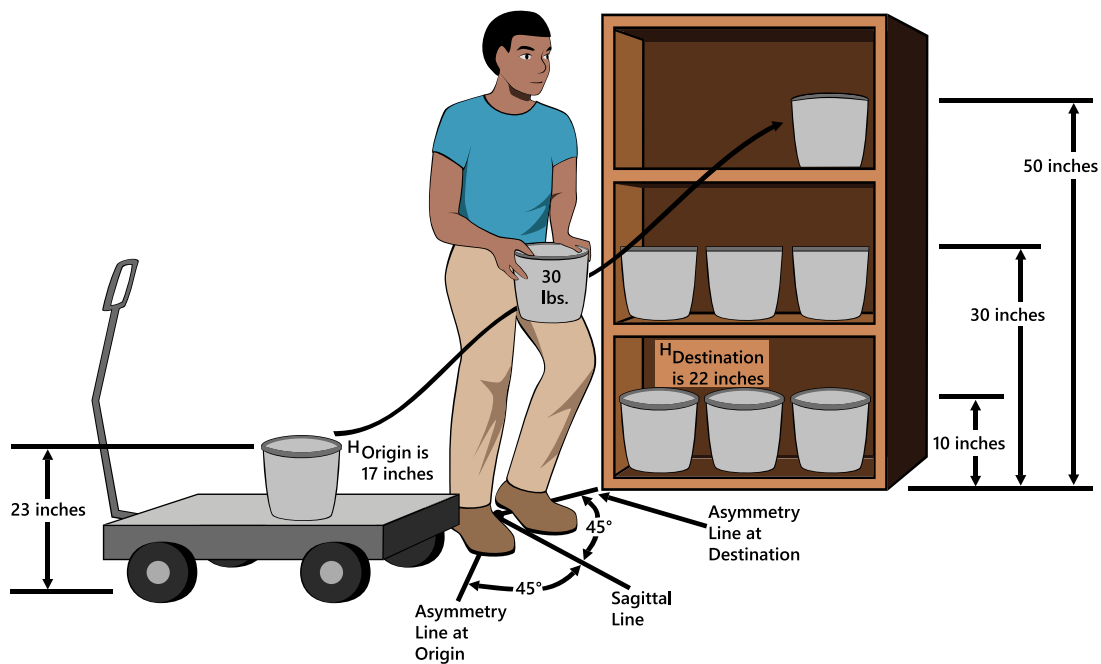


Figure 23: Handling Cans of Liquid, Example 8

This job is divided into three tasks. Task 1 is defined as lifting from the cart to the lower shelf. Task 2 is defined as lifting to the center shelf, and Task 3 is defined as lifting to the upper shelf. Since task 3 requires a reposition of grip at the destination, it must be analyzed at both the origin (Task 3a) and the destination of the Lift (Task 3b). The left and right shelf positions are considered to be equivalent, since the worker can step toward the shelf during the lift.

The following task variable data were measured and recorded on the job analysis worksheet (Figure 24):

1. Cans are 8 inches in height.
2. Cart is 15 inches high.
3. Shelf 1 is 2 inches high.
4. Shelf 2 is 22 inches high.
5. Shelf 3 is 42 inches high.
6. At the origin, the horizontal distance (H) is 17 inches, the vertical height (V) is 23 inches, and the asymmetry angle (A) is 45° for all lifts.
7. At the destination, (H) is 22 inches, and A is 45° for all lifts.
8. The cans are lifted in an intermittent work pattern at a rate of 9 lifts/min (i.e., 3 Lifts/min per shelf) for a duration of 1 hour.
9. Using Table 6, the couplings are classified as poor.

The multi-task lifting analysis consists of the following three steps:

1. Compute the frequency-independent-RWL (FIRWL) and frequency-independent-lifting index FILI values for each task using a default FM of 1.0.
2. Compute the single-task- RWL (STRWL) and single-task-lifting index (STLI) for each task. Note: Since the work pattern is not continuous for the 15-minute sample, the lifting frequency is adjusted using the special procedure.
3. Renumber the tasks in order of decreasing physical stress as determined from the STLI value, starting with the task with the largest STLI.

Step 1

Compute the FIRWL and the FILI values for each task using a default FM of 1.0. The other multipliers are computed from the lifting equation or determined from the multiplier tables (Table 1 to 5, and Table 7). The FIRWL and FILI values are computed only at the origin for Tasks 1 and 2, but since significant control is required for Task 3, the values must be computed at both the origin and destination of the lift.

	FIRWL	FILI
Task 1	21.2 lbs	1.4
Task 2	22.1 lbs	1.4
Task 3a	19.7 lbs	1.5
Task 3b	13.7 lbs	2.2

These results indicate that all of the tasks may *require considerable strength*, especially at the destination of Task 3. Remember, however, that these results do not take the frequency of lifting into consideration.

Step 2

Compute the STRWL and STLI values for each task, where the STRWL for a task is equivalent to the product of the FIRWL and the FM for that task. In this example, the work pattern is intermittent so the frequency is adjusted using the special procedure. Thus, for this job, $F = (3 \text{ lifts/minute} \times 6 \text{ minutes/period} \times 2 \text{ periods}) / 15 \text{ minutes}$, which is equal to 36/15, or 2.4 lifts/minute. As in the previous example, the FM values must be determined by interpolating between the FM values for 2 and 3 lifts/minute from Column 2 of Table 5. The results are displayed in Figure 24 and summarized below.

	STRWL	STLI
Task 1	19.1 lbs	1.6
Task 2	19.9 lbs	1.5
Task 3a	17.7 lbs	1.7
Task 3b	12.4 lbs	2.4

These results indicate that all of the tasks would be particularly stressful, if performed individually. Note, however, that these values do not consider the combined effects of all of the tasks.

Step 3

Renumber the tasks starting with the task with the largest STLI value, and ending with the task with the smallest STLI value. If more than one task has the same STLI value, assign the lower task number to the task with the highest frequency.

3.4.2.3 Hazard Assessment

Compute the composite-lifting index (CLI) using the renumbered tasks. Recall that a special procedure is used to determine the appropriate FM values when (1) repetitive lifting is performed for short durations, and (2) sufficient recovery periods are provided. For example, the frequency for each task in this example is determined by multiplying the actual frequency rate (3 lifts per minute) times the duration (12 minutes), and dividing the result by 15 minutes to obtain an adjusted frequency rate of 2.4 lifts per minute, which is used to compute the CLI.

As shown in Figure 24, the CLI for this job is 2.9, which indicates that there is a significant level of physical stress associated with this job. It appears that strength is a problem for all three tasks, since the FILI values all exceed 1.0. Therefore, the overall physical demands of the job are primarily the result of excessive strength demands, rather than the lifting frequency rate. This may not be the case if the duration exceeds 15 minutes, due to an increase in endurance demands.

3.4.2.4 Redesign Suggestions

The worksheet illustrated in Figure 24 shows that the multipliers with the smallest magnitude (i.e., those providing the greatest penalties) are .46 for the HM for Task 3 at the destination; .59 for the HM for Tasks 1, 2, and 3 at the origin; .85 for the VM for Task 3 at the destination; .86 for the AM for all tasks at the origin and destination; and .90 for the CM for all tasks.

Using Table 8, the following job modifications are suggested:

1. Bring the load closer to the worker to increase HM by reducing the size of the can and/or bringing the load between the workers legs.
2. Reduce the angle of twist to increase AM by moving the origin and destination closer together or further apart.
3. Provide containers with handles or handhold cutouts to increase CM.
4. Raise the origin of the lift to increase VM.

Raising the vertical height at the origin would also decrease the vertical displacement (D), and reduce the angle of twist. Since the size of the H value at the origin depends on the size of the container, the only way to reduce H would be to reduce the container size. An additional benefit of reducing container size is an accompanying reduction in H at the destination for Task 3.

If (1) the height of the cart is increased, (2) twisting is eliminated, and (3) Task 3 is deleted, then the FIRWL for Tasks 1 and 2 would be 27.1 lbs (i.e., $51 \times .59 \times 1.0 \times 1.0 \times 1.0 \times 1.0 \times 0.90$), and the FILI would be reduced from 1.4 to 1.1, which would be acceptable to many more workers than before.

As an alternative an engineering modification could include a design that allows the shelves to either revolve vertically or rotate horizontally for more storage space at the optimum lifting height of 30 inches. This design would eliminate the need to bend or reach while lifting, which is a safer design.

3.4.2.5 Comments

In this example, the cans were not stacked higher than a single can on the cart. The cans, however, could be stacked higher. For a second layer, the vertical height (V) at the origin would be near knuckle height (i.e., about 31 inches). The vertical multiplier (VM) would be increased and the FIRWL would be higher than for lifting from the lowest layer, thus reducing the risk. A third layer, however, may increase the risk of overexertion injury and result in a more stressful job for some workers.

3.5 Repetitive Multi-Task, Long-Duration (>2 hrs)

3.5.1 Product Packaging II, Example 9

3.5.1.1 Job description

Rolls of paper weighing 25 lbs each are pulled off a moving conveyor to work stations where they are wrapped and placed in boxes, as shown in Figure 25. Conveyor delivery allows the roll to slide to the wrapping area, but the roll must be manipulated as it is wrapped. After wrapping, the roll is lifted from the table and placed in a box. The box is closed, secured, and lifted to a pallet. The worker completes this operation once per minute for a continuous duration of 8 hours. The worker does not twist when lifting the rolls of paper. The first lift (from the table to the box) requires significant control at the destination. The second lift (from box to pallet) does not require significant control at the destination.

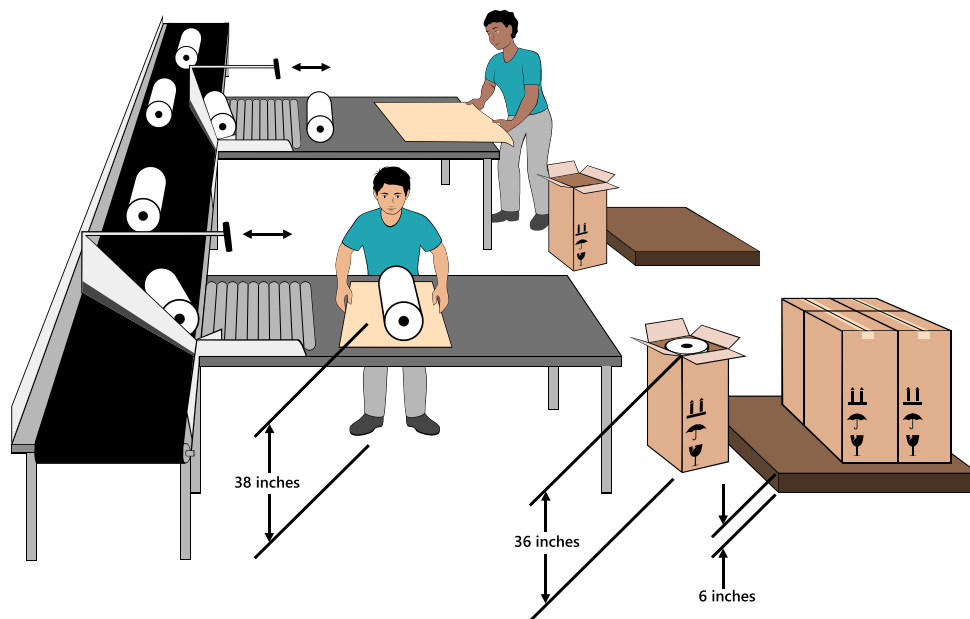


Figure 25: Product Packaging II, Example 9

3.5.1.2 Job Analysis

Since the job consists of more than one task, the multi-task lifting analysis procedure should be used. Task 1 consists of lifting the roll of paper from the table and placing it into a cardboard box, and Task 2 consists of lifting the loaded box from the floor onto the pallet. No asymmetric lifting is involved in either task (i.e., $A = 0$). The following task variable data were measured and recorded on the job analysis worksheet (Figure 26).

Task 1:

1. At the origin of the lift, the horizontal distance (H) is 21 inches and the vertical distance (V) is 38 inches.
2. At the destination of the lift, H is 10 inches and V is 36 inches.
3. If the rolls are handled lengthwise, as shown in Figure 25, then the couplings are classified as “poor”, because the fingers can’t be flexed near 90° (See Table 6).

Task 2:

1. At the origin of the lift, H is 10 inches and V is 0 inches.
2. At the destination of the lift, H is 10 inches and V is 6 inches.
3. The couplings are classified as “fair” because the fingers can be flexed under the box about 90° (See Table 6).

The lifting frequency rate for each task is 1 lift/minute. This means that two lifts occur each minute, since both Task 1 and Task 2 occur about once per minute.

The multi-task lifting analysis consists of the following three steps:

1. Compute the frequency-independent-RWL (FIRWL) and frequency-independent-lifting index FILI values for each task using a default FM of 1.0.
2. Compute the single-task- RWL (STRWL) and single-task-lifting index (STLI) for each task.
3. Renumber the tasks starting with the task with the largest STLI value, and ending with the task with the smallest STLI value. If more than one task has the same STLI value, assign the lower task number to the task with the highest frequency.

Step 1

Compute the FIRWL and the FILI values for each task using a default FM of 1.0. The other multipliers are computed from the lifting equation or determined from the multiplier tables (Tables 1 to 5, and Table 7). Since Task 1 requires significant control at the destination, the FIRWL value must be calculated at both the origin (Task 1a) and the destination (Task 1b) of the lift.

	FIRWL	FILI
Task 1a	20.7 lbs	1.1
Task 1b	44.1 lbs	.6
Task 2	37.8 lbs	.7

The results indicate that these tasks should *not require excessive strength*. Remember, however, that these results do not take the frequency of lifting into consideration.

Step 2

Compute the STRWL and STLI values for each task, where the STRWL for a task is equivalent to the product of the FIRWL and the FM for that task. Based on the given frequencies, vertical heights, and durations, the FM values are determined from Table 5.

The results are displayed in Figure 26 and summarized below.

	STRWL	STLI
Task 1a	15.5 lbs	1.6
Task 1b	33.1 lbs	.8
Task 2	28.4 lbs	.9

These results indicate that, if performed individually, Task 2 would not be stressful, but that Task 1 *would be stressful* for some healthy workers. Note, however, that these values do not consider the combined effects of all of the tasks.

Step 3

Renumber the tasks starting with the task with the largest STLI value, and ending with the task with the smallest STLI value. If more than one task has the same STLI value, assign the lower task number to the task with the highest frequency.

3.5.1.3 Hazard Assessment

Compute the composite-lifting index (CLI) using the renumbered tasks. Only the origin or destination component with the largest STLI is used to compute the CLI for the job when significant control is required for a task. As shown in Figure 26, the CLI for this job is 1.7, which indicates that this job *would be physically stressful for some healthy workers*.

3.5.1.4 Redesign Suggestions

The worksheet illustrated in Figure 26 shows that the multipliers with the smallest magnitude (i.e., those providing the greatest penalties) for this task are .48 for the HM at the origin of Task 1, .78 for the VM for Task 2, and .90 for the CM at the origin and destination of Task 1. Using Table 8, the following job modifications are suggested:

1. Bring the load closer to the worker to increase HM by reducing the size of the roll and/or bringing the load between the workers legs at the origin for Task 1.
2. Raise the vertical height of the lift for Task 2 at the origin and at the destination to increase VM.
3. Provide better couplings for Task 1 to increase CM.

The largest penalty comes from lifting the rolls from the wrapping table into the box. A practical job redesign would be to provide a recess for the box at the end of the table, so that the worker could easily slide the roll into the box without lifting it. The worker could then slide the box to the edge of the table, and lift it from the table to the pallet. This job

modification would allow the worker to get closer to the load when lifting, which would increase the FIRWL and decrease the FILI.

As an alternative job modification, the worker could be rotated from this job to a job with light work every one to two hours to decrease the lifting duration. This would provide a sufficient recovery period for the worker, so that fatigue would not become a problem. The light duty work, however, should last for at least .3 times the amount of time spent on the packaging job.

3.5.1.5 Comments

There is an inherent danger in trying to simplify a complex lifting job. The overriding concern is that the worker is not exposed to excessive biomechanical or physiological stress. This multi-task analysis procedure was designed to provide a series of intermediate values that would help guide the redesign of physically demanding lifting tasks. These values include the FIRWL, FILI, STRWL, and STLI. These intermediate values should not be used as design limits, since they only provide task specific information. The overall risk of injury for a lifting job is dependent upon the combined effects of the job, rather than the individual effects of the tasks.

3.5.2 Warehouse Order Filling, Example 10

3.5.2.1 Job Description

A worker lifts cartons of various sizes from supply shelves onto a cart as illustrated in Figure 27. There are three box sizes (i.e., A, B, and C) of various weights. These lifting tasks are typical in warehousing, shipping, and receiving activities in which loads of

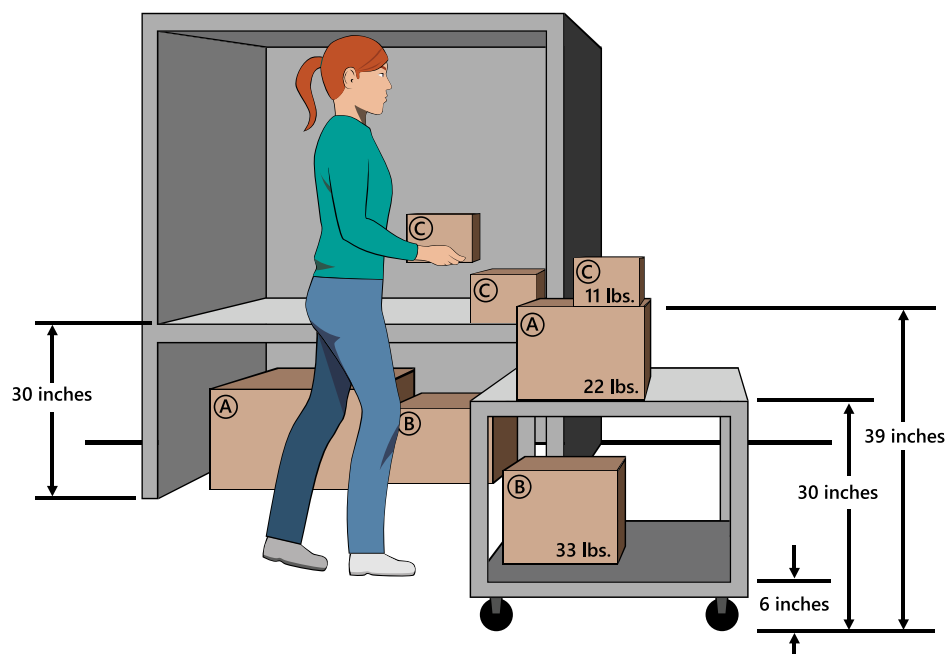


Figure 27: Warehouse Order Filling, Example 10

varying weights and sizes are lifted at different frequencies. Assume that the following observations were made: (1) control of the load is not required at the destination of any lift; (2) the worker does not twist when picking up and putting down the cartons; (3) the worker can get close to each carton; and, (4) walking and carrying are minimized by keeping the cart close to the shelves.

3.5.2.2 Job Analysis

Since the job consists of more than one distinct task and the task variables often change, the multi-task lifting analysis procedure should be used.

This job can be divided into three tasks represented by cartons A, B, and C. The following measurements were made and recorded on the job analysis worksheet figure 28):

1. The horizontal locations (H) for each task at the origin and destination are as follows: Box A, 16 inches; Box B, 12 inches; and Box C, 8 inches.
2. The vertical locations (V) at the origin are taken to be the position of the hands under the cartons as follows: Box A, 0 inches; Box B, 0 inches; and Box C, 30 inches.
3. The vertical locations (V) at the destination are the vertical position on the cart as follows: Box A, 30 inches; Box B, 6 inches; and, Box C, 39 inches.
4. The average weights lifted for each task are as follows: Box A, 22 lbs; Box B, 33 lbs; and, Box C, 11 lbs.
5. The maximum weights lifted for each task are as follows: Box A, 33 lbs; Box B, 44 lbs; and, Box C, 22 lbs.
6. No asymmetric lifting is involved (i.e., $A = 0$).
7. The lifting frequency rates for each task are as follows: Box A, 1 lift/min Box B 2 lifts/min; and Box C 5 lifts/min.
8. The lifting duration for the job is 8 hours, however, the maximum weights are lifted infrequently (i.e., less than or equal to once every five minutes for 8 hours).
9. Using Table 6, the couplings are classified as fair.

The multi-task lifting analysis consists of the following three steps:

1. Compute the frequency-independent-RWL (FIRWL) and frequency-independent-lifting index FILI values for each task using a default FM of 1.0.
2. Compute the single-task-RWL (STRWL) and single-task-lifting index (STLI) for each task.
3. Renumber the tasks in order of decreasing physical stress, as determined by the STLI value, starting with the task with the largest STLI.

Step 1

Compute the FIRWL and the FILI values for each task using a default FM of 1.0. The other multipliers are computed from the lifting equation or determined from the multiplier tables (Tables 1 to 5, and Table 7). Recall that the FILI is computed for each task by dividing the *maximum* weight of that task by its FIRWL.

	FIRWL	FILI
Task 1	21.0 lbs	1.6
Task 2	31.4 lbs	1.4
Task 3	51.0 lbs	.4

These results indicate that two of the tasks require strength demands that exceed the RWL level. Remember, however, that these results do not take the frequency of lifting into consideration.

Step 2

Compute the STRWL and STLI values for each task, where the STRWL for a task is equivalent to the product of the FIRWL and the FM for that task. Recall that the STLI is computed for each task by dividing the *average* weight of that task by its STRWL. The appropriate FM values are determined from Table 5.

	STRWL	STLI
Task 1	15.8 lbs	1.4
Task 2	20.4 lbs	1.6
Task 3	17.8 lbs	.6

These results indicate that Tasks 1 and 2 would be stressful for some workers, if performed individually. Note, however, that these values do not consider the combined effects of all of the tasks.

Step 3

Renumber the tasks in order of decreasing physical stress, as determined by the STLI value, starting with the task with the highest STLI value.

3.5.2.3 Hazard Assessment

Compute the composite-lifting index (CLI) using the renumbered tasks. As shown in Figure 28, the CLI for this job is 3.6, which indicates that this job would be physically stressful for nearly all workers. Analysis of the results suggests that the combined effects of the tasks are significantly more stressful than any individual task.

3.5.2.4 Redesign Suggestions

Developing a redesign strategy for a job depends on tangible and intangible factors that may be difficult to evaluate, including costs/benefits, feasibility, and practicality. No preferred procedure has been developed and tested. Therefore, the following suggestions represent only one approach to ergonomic job modification.

In this example the magnitude of the FILI, STLI and CLI values indicate that both strength and endurance would be a problem for many workers. Therefore, the redesign should attempt to decrease the physical demands by modifying the job layout and decrease the physiological demands by reducing the frequency rate or duration of continuous lifting. If the maximum weights were eliminated from the job, then the CLI

would be significantly reduced, the job would be less stressful, and more workers could perform the job than before.

Those lifts with strength problems should be evaluated for specific engineering changes, such as (1) decreasing carton size or removing barriers to reduce the horizontal distance; (2) raising or lowering the origin of the lift; (3) reducing the vertical distance of the lift; improving carton couplings, and (4) decreasing the weight to be lifted. The redesign priority for this example is based on identifying interventions that provide the largest increase in the FIRWL for each task (Step 2 on worksheet). For example, the maximum weight lifted for carton A is unacceptable; however, if the carton at the origin were on the upper shelf, then the FIRWL for Task 1 would increase from 21.0 lbs to 27.0 lbs. The maximum weight lifted still exceed the FIRWL, but lifts of average weight are now below the FIRWL. Additionally, providing handles, decreasing box size, or reducing the load to be lifted will decrease the stress of manual lifting.

3.5.2.5 Comments

This example demonstrates the complexity of analyzing multi-task lifting jobs. Errors resulting from averaging and errors introduced by ignoring other factors (e.g., walking, carrying, holding, pushing and pulling activities, and environmental stressors), can only be resolved with detailed biomechanical, metabolic, cardiovascular, and psychophysical evaluations.

Several important application principles are illustrated in this example:

1. The horizontal distance (H) for Task 3 was less than the 10.0 inches minimum. Therefore, H was set equal to 10 inches (i.e., multipliers must be less than or equal to 1.0).
2. The vertical travel distance (D) in Task 2 was less than the 10 inches minimum. Therefore, D was set equal to 10 inches.